AN AFFINE SCHEME AND AN AFFINE OPEN WHOSE COMPLEMENT ADMITS NO QUASI-COMPACT DENSE OPEN

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The statement's item 1 is witnessed by (T, U) and item 2 by (S, s); the (T, t) = (S, s) isomorphism used later in the chapter is a separate remark, not needed here.

Editorial note. The label lemma-complement-of-affine-does-not-contain-qc-dense-open names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 2249-2261: A = k[z_1, z_2, ...]/(all pairwise products z_i z_j, i != j), S = Spec(A), s the closed point of the maximal ideal (z_i); S - {s} = union of the D(z_i), each D(z_i) open irreducible with generic point eta_i; any quasi-compact open V of S - {s} dense in S - {s} would contain all eta_i, but a principal affine open D(f) with f in (z_1, z_2, ...) lies in (z_1, ..., z_n) for some n and contains only finitely many eta_i, so V cannot be quasi-compact
- Context span 2263-2274: B = k[x, z_1, z_2, ...]/(x z_i for all i and all pairwise products z_i z_j), T = Spec(B), U = D(x); for t the closed point of (x, z_i) the closure of U is U cup {t}; a quasi-compact open V with V cap U = empty and U cup V dense in T would be a quasi-compact open of union_i D(z_i) dense in union_i D(z_i), which the first paragraph excludes

Let k be a field and let $A = k[z_1, z_2, z_3, \ldots]/I$ where I is the ideal generated by all pairwise products $z_i z_j$, $i \neq j$, $i, j \in \mathbf{N}$. Set $S = \mathrm{Spec}(A)$. Let $s \in S$ be the closed point corresponding to the maximal ideal (z_i). We claim there is no quasi-compact open $V \subset S \setminus \{s\}$ which is dense in $S \setminus \{s\}$. Note that $S \setminus \{s\} = \bigcup D(z_i)$. Each $D(z_i)$ is open and irreducible with generic point η_i. We conclude that $\eta_i \in V$ for all i. However, a principal affine open of $S \setminus \{s\}$ is of the form $D(f)$ where $f \in (z_1, z_2, \ldots)$. Then $f \in (z_1, \ldots, z_n)$ for some n and we see that $D(f)$ contains only finitely many of the points η_i. Thus V cannot be quasi-compact.

Let k be a field and let $B = k[x, z_1, z_2, z_3, \ldots]/J$ where J is the ideal generated by the products xz_i, $i \in \mathbf{N}$ and by all pairwise products $z_i z_j$, $i \neq j$, $i, j \in \mathbf{N}$. Set $T = \mathrm{Spec}(B)$. Consider the principal open $U = D(x)$. We claim there is no quasi-compact open $V \subset T$ such that $V \cap U = \emptyset$ and $V \cup U$ is dense in T. Let $t \in T$ be the closed point corresponding to the maximal ideal (x, z_i). The closure of U in T is $\overline{U} = U \cup \{t\}$. Hence $V \subset \bigcup_i D(z_i)$ is a quasi-compact open. By the arguments of the previous paragraph we see that V cannot be dense in $\bigcup D(z_i)$.

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Lemma 0.1. *Nonexistence quasi-compact opens of affines:*

(1) *There exist an affine scheme S and affine open $U \subset S$ such that there is no quasi-compact open $V \subset S$ with $U \cap V = \emptyset$ and $U \cup V$ dense in S.*
(2) *There exists an affine scheme S and a closed point $s \in S$ such that $S \setminus \{s\}$ does not contain a quasi-compact dense open.*

Proof. See discussion above. □